

Alternating-zero local multiplicities in odd-power cyclic exchange relations

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Abstract

For an odd integer $k \geq 3$, consider the finite cyclic relation algebra defined over $\mathbb{C}$ by $x_i^k + 1 = x_{i-1}x_{i+1}$, with indices modulo n . We determine its complete local Artin length at every alternating-zero point, for every even clock $n = 2m$. These points have a common local model given by the logarithmic gradient of a formal dilogarithm potential. Intersection additivity expresses its length as a sum over proper cyclic subsets with run weights 1, $k - 1$, and $2k - 1$, together with a full-cycle term equal to 1 unless $4 \mid m$, when it is $4k + 1$. The proof resolves the one-dimensional cancelled path kernel and the two-dimensional cycle kernel, including the exceptional cubic coefficients. If $\ell_m(k)$ is the resulting local length and $D_k(t) = 1 - t - (k - 1)t^2 - t^3 - 2kt^4$, then

$$\exp\left(\sum_{m \geq 1} \frac{\ell_m(k)}{m} t^m\right) = \frac{1}{(1 - t)D_k(t)(1 - t^4)^{k-1}}.$$

The alternating support is already known from rank-two cluster geometry; the calculation here concerns its nonreduced thickness in the unsaturated cyclic relation algebra. It neither classifies the other boundary supports nor identifies the displayed series with an ordinary dynamical zeta function.

1 Introduction

Clearing the denominators in a rational recurrence produces polynomial relations that remain meaningful at zero coordinates. Their solutions may carry nilpotent structure that is invisible in the recurrence's original domain. Even when the possible zero patterns are understood, determining the lengths of the corresponding local algebras is a separate intersection problem. This distinction is particularly concrete for the exchange recurrence

$$x_{i-1}x_{i+1} = 1 + x_i^k, \quad k \geq 3 \text{ odd.} \quad (1.1)$$

Imposing a cyclic clock gives a finite algebra. At an alternating-zero point its local embedding dimension grows with the clock, so a description of the support on a two-dimensional cluster variety cannot by itself determine the retained infinitesimal structure.

The support in question is classical in this setting. Beyer and Muller classify the deep points of coefficient-free rank-two cluster algebras; their alternating pattern is $(0, \alpha, 0, \alpha^{-1})$, with the relevant root condition, and the shifted phase [1, Section 4, especially Corollary 4.6 in arXiv v1]. We use this attribution while deriving the elementary finite-clock restriction directly. Benito, Faber, Mourtada, and Schöber classify the ambient rank-two singularities; their Jacobian description implies smoothness for the positive rank-two exponents over $\mathbb{C}$ [2, Section 5, Theorem 5.2.3 in arXiv v2]. The smooth ambient surface and the unsaturated cyclic relation algebra studied here are different schemes. A further related, but distinct, direction is the birational dynamics of cluster mutation maps studied by Grigorev, Kalidindi, Quintero

Santander, and Roeder [3]. For equal exponents their mutation composition corresponds to the square of $F_k(x, y) = (y, (1 + y^k)/x)$. We do not use a Picard-action or entropy calculation to infer a boundary length.

Our result determines the local thickness of the entire alternating-zero stratum, simultaneously for every odd $k \geq 3$ and every clock at which it occurs. The local equations reduce to products of coordinates and gradient components of one formal potential. Expanding their intersection multiplicity over all factor choices leaves path and cycle gradient colengths. Nonsingular Hessians account for the simplest terms, but they do not settle the problem: paths of odd length have a one-dimensional kernel, and cycles of length divisible by four have a two-dimensional kernel. The cancelled path case has leading effective degree $2k$ rather than k . The cycle case requires both a mixed quartic coefficient and a pure-axis coefficient. Both calculations have additional terms at $k = 3$.

The resulting weights depend only on path length modulo four. This makes the local lengths amenable to a labeled cyclic-word enumeration and yields a rational multiplicity-generating function. The rational series is a consequence of the same local intersection calculation, not a new dynamical realization. The original rational recurrence excludes these zero-coordinate points from its admissible torus domain, and other boundary supports remain outside the theorem.

Section 2 states the formula and fixes the small-clock conventions. Section 3 proves the formal reduction and the required nonreduced intersection additivity. The path and cycle calculations occupy sections 4 and 5. We derive the rational series in section 6, and distinguish the finite computational controls from the proof and the unresolved boundary problem in section 7.

2 Cyclic relation algebras and the main theorem

Fix an odd integer $k \geq 3$. For $n \geq 1$ define

$$\mathcal{A}_{k,n} = \mathbb{C}[x_0, \dots, x_{n-1}] / (x_i^k + 1 - x_{i-1}x_{i+1} : 0 \leq i < n), \quad (2.1)$$

where indices are modulo n . The neighbor product is literal: at $n = 2$ the equation for x_i contains the square of the other coordinate. This is the *unsaturated cyclic relation algebra*. No saturation by the coordinates, cluster-surface replacement, or passage to reduced points is part of its definition.

Lemma 2.1. *The algebra $\mathcal{A}_{k,n}$ has dimension k^n over $\mathbb{C}$. For $n = 2m$, its alternating-zero points in one phase are*

$$x_{2j} = 0, \quad x_{2j+1} = t_j, \quad t_j^k = -1, \quad t_{j-1}t_j = 1 \quad (j \bmod m). \quad (2.2)$$

There are k choices in each phase when m is even, and one choice in each phase when m is odd. The two phases are disjoint.

Proof. For a monomial order refining total degree, the displayed relations have leading monomials x_i^k , since $k > 2$. These monomials are pairwise relatively prime, so the relations form a Gröbner basis by the product criterion. The standard monomials have all exponents below k ; there are k^n of them. Thus every localization at a point is an Artin local algebra.

Substitution of alternating zeros gives exactly (2.2). The product conditions imply $t_j = t_0^{(-1)^j}$. If m is even this is cyclic for any of the k roots of -1 . If m is odd, closure forces $t_0^2 = 1$; for odd k , the condition $t_0^k = -1$ leaves only $t_0 = -1$. Shifting all indices by one gives the other phase. Its zero coordinates and nonzero coordinates are interchanged, hence the phases cannot coincide. $\square$

These are labeled solutions of the cyclic equations, not a count of orbits of exact period $2m$. In particular the support pattern has period dividing four. We shall prove that all its permitted

root choices and both phases have isomorphic local algebras. Denote their common length by $\ell_m(k)$.

For combinatorial purposes let C_m be the labeled cyclic set of m sites. Its cycle potential has one loop when $m = 1$ and two copies of the edge when $m = 2$. Every nonempty proper induced connected component is an ordinary path, with an isolated site counted as a path of length one. For a proper subset $S \subsetneq C_m$, let $\text{Runs}(S)$ be its maximal cyclic runs of selected sites. Put

$$w_L(k) = \begin{cases} 1, & L \equiv 0, 2 \pmod{4}, \\ k-1, & L \equiv 1 \pmod{4}, \\ 2k-1, & L \equiv 3 \pmod{4}, \end{cases} \quad c_m(k) = \begin{cases} 1, & 4 \nmid m, \\ 4k+1, & 4 \mid m. \end{cases} \quad (2.3)$$

Theorem 2.2 (Alternating-zero local length). *For every odd $k \geq 3$ and every $m \geq 1$, all alternating-zero points of $\mathcal{A}_{k,2m}$ have isomorphic local algebras, of length*

$$\ell_m(k) = c_m(k) + \sum_{S \subsetneq C_m} \prod_{R \in \text{Runs}(S)} w_{|R|}(k). \quad (2.4)$$

The empty subset contributes one. Their total contribution to the length of $\mathcal{A}_{k,2m}$ is $2\ell_m(k)$ when m is odd and $2k\ell_m(k)$ when m is even.

Corollary 2.3 (Rational multiplicity series). *Define*

$$D_k(t) = 1 - t - (k-1)t^2 - t^3 - 2kt^4. \quad (2.5)$$

As an identity of formal power series,

$$Z_k(t) := \exp\left(\sum_{m \geq 1} \frac{\ell_m(k)}{m} t^m\right) = \frac{1}{(1-t)D_k(t)(1-t^4)^{k-1}}. \quad (2.6)$$

Let b_n be the total alternating-zero contribution at clock n , with $b_n = 0$ for odd n . Then

$$\exp\left(\sum_{n \geq 1} \frac{b_n}{n} u^n\right) = Z_k(u^2)^{(k+1)/2} Z_k(-u^2)^{(k-1)/2}. \quad (2.7)$$

The theorem is proved by the local reductions in the next three sections; the corollary is proved by an exact double count in section 6. For orientation, the first four lengths are

$$\ell_1 = 2, \quad \ell_2 = 2k, \quad \ell_3 = 3k+2, \quad \ell_4 = 2k(k+6). \quad (2.8)$$

The weights in (2.3) are local gradient colengths, not numbers of reduced branches. This is why the intersection argument below must retain nilpotents.

3 Formal reduction and intersection additivity

All series in this section are formal series over $\mathbb{C}$. Write

$$\text{Li}_2(s) = \sum_{q \geq 1} \frac{s^q}{q^2}.$$

We use this defining series only on elements of a maximal ideal; no analytic continuation or global dilogarithm identity is required.

Proposition 3.1 (Common local presentation). *At any point (2.2), the completed local algebra is*

$$\mathbb{C}[[z_0, \dots, z_{m-1}]] / (z_j \partial_j V_m : 0 \leq j < m), \quad V_m(z) = \sum_{j=0}^{m-1} \text{Li}_2(z_j z_{j+1}) - \sum_{j=0}^{m-1} \text{Li}_2(-z_j^k). \quad (3.1)$$

The formula uses cyclic indices and counts every summand literally. In particular its local embedding dimension is m .

Proof. Set $z_j = x_{2j}$ and $b_j = x_{2j+1}$. The odd-index equations are $b_j^k = z_j z_{j+1} - 1$. Their derivatives $k b_j^{k-1}$ are units at the chosen point, so the formal implicit-function theorem eliminates them as

$$b_j = t_j (1 - z_j z_{j+1})^{1/k}. \quad (3.2)$$

Here the fractional power is its unique expansion with constant term one. The remaining equations are $b_{j-1} b_j = 1 + z_j^k$. By $t_{j-1} t_j = 1$, their unit ratios have constant term one and logarithms

$$\frac{1}{k} \log(1 - z_{j-1} z_j) + \frac{1}{k} \log(1 - z_j z_{j+1}) - \log(1 + z_j^k).$$

For a unit U with constant term one, $\log U = (U - 1)h(U - 1)$ with $h(0) = 1$. Thus taking this logarithm replaces each equation by a unit multiple, not merely an equation with the same zero set. After multiplication by k , the remaining ideal has generators

$$L_j = \log(1 - z_{j-1} z_j) + \log(1 - z_j z_{j+1}) - k \log(1 + z_j^k). \quad (3.3)$$

Direct differentiation of the defining series gives $L_j = -z_j \partial_j V_m$. If $m = 1$, differentiating the loop $\text{Li}_2(z_0^2)$ contributes twice; if $m = 2$, the two copies of $\text{Li}_2(z_0 z_1)$ each contribute. These are exactly the duplicated neighbor terms in (3.3).

The presentation is independent of every t_j . A one-step cyclic shift identifies the other phase. Since the original local algebra is Artin by lemma 2.1, completion does not change it or its length. Finally V_m has no constant or linear term, so every $z_j \partial_j V_m$ belongs to the square of the maximal ideal. There are therefore no linear relations among the m generators of its cotangent space, which proves the embedding-dimension assertion. $\square$

Remark 3.2 (The ambient-scheme distinction). Any closed subscheme of a smooth surface has local embedding dimension at most two: its cotangent space is a quotient of the surface's cotangent space. For $m \geq 3$, proposition 3.1 consequently rules out an identification of the local algebra here with the fixed-point local scheme of an endomorphism of a smooth cluster surface. The smoothness statement for the relevant rank-two cluster surface is an existing result [2, Theorem 5.2.3 in arXiv v2]; it is not a conclusion about (2.1). The support's cluster interpretation does not identify these nonreduced structures.

The next lemma supplies the particular intersection-additivity fact that we need, with no reducedness hypothesis.

Lemma 3.3 (Splitting product generators). *Let $R = \mathbb{C}[[z_1, \dots, z_d]]$, and let a_i, b_i lie in its maximal ideal. Suppose every ideal generated by choosing one of a_i, b_i for each i has finite colength. Then*

$$\text{length } R / (a_1 b_1, \dots, a_d b_d) = \sum_{\varepsilon \in \{0,1\}^d} \text{length } R / (c_1^{\varepsilon_1}, \dots, c_d^{\varepsilon_d}), \quad c_i^0 = a_i, \quad c_i^1 = b_i. \quad (3.4)$$

The symbols c_i^0, c_i^1 in this formula denote the two choices, not powers.

Proof. Every intermediate ideal with some product generators and some chosen factors also has finite colength. Indeed, a prime containing such an ideal contains at least one factor from each remaining product, and hence contains one of the assumed finite-colength ideals. It must therefore be the maximal ideal.

Fix $d-1$ intermediate generators and let Q be their quotient ring. Adjoining the last product, or either of its factors a, b , gives a system of parameters in the regular local ring R . A regular local ring is Cohen–Macaulay, and any system of parameters is a regular sequence. Thus Q is a one-dimensional Cohen–Macaulay ring, and a, b are parameters and nonzerodivisors in Q . There is an exact sequence

$$0 \longrightarrow Q/(a) \xrightarrow{\times b} Q/(ab) \longrightarrow Q/(b) \longrightarrow 0. \quad (3.5)$$

Injectivity follows by cancelling the nonzerodivisor b ; the other exactness statements follow from the quotient maps. Lengths in (3.5) add. Splitting the product generators successively proves (3.4). None of these rings is required to be reduced. $\square$

For a formal potential P with an isolated critical point at the origin, write

$$\mu(P) = \dim_{\mathbb{C}} \mathbb{C}[[z]]/(\partial_1 P, \dots, \partial_d P)$$

for its gradient colength. On a path of L sites the restricted potential is

$$P_L(z) = \sum_{j=1}^{L-1} \text{Li}_2(z_j z_{j+1}) - \sum_{j=1}^L \text{Li}_2(-z_j^k). \quad (3.6)$$

If $S \subsetneq C_m$, setting all unselected coordinates to zero splits $V_m|_S$ as the sum of the path potentials on $\text{Runs}(S)$. Choosing $\partial_j V_m$ for $j \in S$ and z_j for $j \notin S$ in lemma 3.3 gives precisely this restricted gradient ideal. The disjoint components involve disjoint sets of variables, so their finite local algebras tensor over $\mathbb{C}$ and their lengths multiply. It follows, provided the gradient colengths are finite, that

$$\ell_m(k) = \mu(V_m) + \sum_{S \subsetneq C_m} \prod_{R \in \text{Runs}(S)} \mu(P|_R). \quad (3.7)$$

Finiteness and the exact values of these gradient colengths are established in sections 4 and 5, so this application will not assume what it needs to prove.

We record also the elimination estimate used in both calculations. The order of a vector-valued series is the least total degree of any nonzero component; $O(a^s)$ means that every term has total degree at least s .

Lemma 3.4 (Eliminating a nonsingular Hessian block). *Let $P(a, u)$ be a formal series with $dP(0) = 0$ whose quadratic part is $\frac{1}{2}u^T H u$, with H invertible. There is a unique series $u = u(a)$ vanishing at the origin that solves $\partial_u P = 0$, and the full gradient algebra is isomorphic to the gradient algebra of $p(a) = P(a, u(a))$. If an approximation $u_0(a)$ vanishes at the origin and satisfies $\partial_u P(a, u_0(a)) = O(a^s)$ for $s \geq 2$, then*

$$u(a) - u_0(a) = O(a^s), \quad P(a, u(a)) - P(a, u_0(a)) = O(a^{2s}). \quad (3.8)$$

Proof. The derivative of $\partial_u P$ with respect to u at the origin is H . Formal implicit elimination gives the unique $u(a)$, and quotienting by the complementary equations is substitution of that solution. The chain rule gives $\partial_a p = (\partial_a P)(a, u(a))$, since the complementary gradient is zero there; hence the remaining ideals agree. Subtract the complementary equations at u_0 and u and compare their lowest degrees. The invertible linear coefficient H shows that their solution difference has order at least s . Taylor expansion of P at u_0 now has a linear term of order at least $s + s$ and quadratic and higher difference terms of order at least $2s$. This proves (3.8). $\square$

4 Path gradient colengths

Proposition 4.1. *For every $L \geq 1$, the gradient ideal of P_L in (3.6) has finite colength*

$$\mu(P_L) = w_L(k).$$

4.1 The Hessian and the uncanceled kernel

The quadratic part of P_L is $\sum_{j=1}^{L-1} z_j z_{j+1}$. Its Hessian H_L is the ordinary path adjacency matrix. Expanding its determinant at an endpoint gives $\det H_L = -\det H_{L-2}$, with $\det H_0 = 1$ and $\det H_1 = 0$. Consequently H_L is invertible for even L . Formal implicit elimination of the full gradient then gives $\mu(P_L) = 1$.

For odd L , solving $H_L v = 0$ recursively from an endpoint shows that the kernel is the one-dimensional space generated by

$$v = (1, 0, -1, 0, 1, 0, -1, \dots). \quad (4.1)$$

For the bilinear dot product $v^T v = (L+1)/2 \neq 0$. Thus $\mathbb{C}^L = \text{span}(v) \oplus v^\perp$, and the symmetric Hessian is invertible on $v^\perp$. Write $z = av + u$ with $u \in v^\perp$ and solve the complementary gradient equations for $u = u(a)$. By lemma 3.4, the gradient colength equals the colength of $(p'_L(a))$, where $p_L(a) = P_L(av + u(a))$.

At av , every adjacent product is zero. The nonlinear gradient starts in degree $k-1$, so $u(a) = O(a^{k-1})$, and elimination changes the restricted potential $P_L(av)$ only in degree at least $2k-2$. If $L = 4r+1$, the nonzero entries of v have one more $+1$ than -1 . Since k is odd and $-\text{Li}_2(-s) = s + O(s^2)$,

$$p_L(a) = a^k + O(a^{k+1}). \quad (4.2)$$

Here $2k-2 > k$ for every $k \geq 3$, so complementary elimination cannot change the displayed coefficient. Therefore p'_L has order $k-1$, and $\mu(P_L) = k-1$. This also proves the isolated-site case $L = 1$.

4.2 The cancelled kernel at lengths three modulo four

It remains to treat $L = 4r-1$, $r \geq 1$. Reflection of the path sends v to $-v$ and preserves $v^\perp$. It preserves the potential, so the uniqueness of the complementary solution implies $p_L(-a) = p_L(a)$. The degree- k restriction vanishes, since the $2r$ nonzero entries of v consist of equal numbers of the two signs.

Set $q = k-1$, which is even. The leading nonlinear gradient at av is ka^q at every odd site and zero at every even site. Its projection onto the kernel is zero. The first complementary correction that solves the Hessian equation is

$$(u_0)_{2j-1} = 0, \quad (u_0)_{2j} = \begin{cases} -ka^q, & j \text{ odd}, \\ 0, & j \text{ even}. \end{cases} \quad (4.3)$$

Indeed, each odd site's adjacent even entries sum to $-ka^q$; the missing entries at the two endpoints are zero. Each even site's adjacent odd correction entries sum to zero. There are exactly r nonzero even entries, and u_0 is orthogonal to v .

We next bound the residual rather than assuming this first correction is exact. The higher pair-potential gradient at an even site begins with $u_{\text{even}} a^2$, of degree $k+1$. At an odd site it begins with au_{even}^2 , of degree $2k-1$. The unary derivative at a corrected even site has degree q^2 , and the higher unary derivatives at odd sites start in degree $2k-1$. For odd $k \geq 3$,

$$2k-1 \geq k+1, \quad (k-1)^2 \geq k+1; \quad (4.4)$$

the second inequality is an equality at $k = 3$. The linear Hessian contribution cancels the leading unary gradient by (4.3). Hence the full gradient residual, and in particular its complementary part, has order at least $k + 1$. Lemma 3.4 gives

$$u(a) - u_0(a) = O(a^{k+1}), \quad p_L(a) - P_L(av + u_0(a)) = O(a^{2k+2}). \quad (4.5)$$

It therefore suffices to evaluate the displayed approximation through degree $2k$.

Each nonzero even entry lies between $+a$ and $-a$, and the corresponding pair terms satisfy the formal identity

$$\text{Li}_2(-ka^k) + \text{Li}_2(ka^k) = \frac{1}{2} \text{Li}_2(k^2 a^{2k}). \quad (4.6)$$

This identity follows by cancelling the odd powers in the defining series. With r such entries, the total pair coefficient in degree $2k$ is $rk^2/2$. Opposite odd sites give

$$-\text{Li}_2(-a^k) - \text{Li}_2(a^k) = -\frac{1}{2} \text{Li}_2(a^{2k});$$

their r pairs contribute $-r/2$ in that degree. The r corrected even sites each have leading unary term $(-ka^q)^k$. Its degree is $k(k-1)$, which exceeds $2k$ if $k \geq 5$. At $k = 3$ it instead contributes $-27a^6$ per corrected site. Thus

$$p_{4r-1}(a) = \begin{cases} \frac{r(k^2-1)}{2} a^{2k} + O(a^{2k+1}), & k \geq 5 \text{ odd}, \\ -23ra^6 + O(a^7), & k = 3. \end{cases} \quad (4.7)$$

No lower-degree term survives: the paired evaluations start in degree $2k$, the corrected unary degree is at least $2k$, and the elimination error starts in degree $2k+2$. Both displayed coefficients are nonzero over $\mathbb{C}$. The derivative therefore has order $2k-1$ and finite colength $2k-1$. Together with the previous cases, this proves proposition 4.1.

5 Cycle gradient colengths

Proposition 5.1. *For every $m \geq 1$, the gradient ideal of the cycle potential V_m has finite colength $\mu(V_m) = c_m(k)$.*

5.1 The Hessian and period-four reduction

The Hessian of the quadratic part $\sum_j z_j z_{j+1}$ acts by

$$(H_m z)_j = z_{j-1} + z_{j+1}.$$

This also gives $H_1 = [2]$ and the off-diagonal entries 2 in H_2 ; these are the loop and double-edge conventions of section 2. The cyclic Fourier vectors have eigenvalues $2 \cos(2\pi j/m)$, $0 \leq j < m$, so H_m is invertible if and only if $4 \nmid m$. In these cases the full-gradient implicit-function theorem gives colength one.

Let $m = 4r$. The kernel is the two-dimensional space of repeated patterns

$$z = (a, b, -a, -b, a, b, -a, -b, \dots). \quad (5.1)$$

The two displayed kernel basis vectors are orthogonal, each with squared length $2r$, so the Hessian is invertible on their orthogonal complement. Let $p_m(a, b)$ be the potential after eliminating that complement. Shift by four preserves the potential, the kernel point, and the chosen complement. Uniqueness of formal elimination forces its solution to be period four. On the period-four subspace $V_m = rV_4$, and the complementary equations restrict to r times the four-site equations. Consequently

$$p_m(a, b) = rp_4(a, b). \quad (5.2)$$

Multiplication by the nonzero scalar r does not change a gradient ideal over $\mathbb{C}$.

Two reflections of the four-cycle induce $(a, b) \mapsto (-a, b)$ and $(a, b) \mapsto (a, -b)$ on the kernel. They preserve the orthogonal complement, so uniqueness again makes the effective potential invariant. A one-step shift exchanges a, b up to signs. We obtain the exact formal description

$$p_4(a, b) = W(A, B), \quad A = a^2, \quad B = b^2, \quad W(A, B) = W(B, A). \quad (5.3)$$

These are graph symmetries; no global sign symmetry of the odd-degree unary potential is assumed.

We shall compute the mixed quartic coefficient and the first pure-axis coefficient:

$$[a^2 b^2] p_4 = \alpha, \quad p_4(a, 0) = \beta a^{2k} + O(a^{2k+2}), \quad (5.4)$$

where

$$(\alpha, \beta) = \begin{cases} (1, (k^2 - 2)/4), & k \geq 5 \text{ odd}, \\ (-8, -5), & k = 3. \end{cases} \quad (5.5)$$

In particular both coefficients are nonzero, and there is no earlier pure-axis term.

5.2 The mixed quartic coefficient

On the kernel $(a, b, -a, -b)$, the four adjacent products alternate between ab and $-ab$. Their pair potential is

$$2 \operatorname{Li}_2(ab) + 2 \operatorname{Li}_2(-ab) = \operatorname{Li}_2(a^2 b^2),$$

whose mixed quartic coefficient is one. For $k \geq 5$, the unary potential has degree at least five. Complementary corrections start in degree at least three (the pair nonlinearity is cubic in the gradient), and so their effect on the potential starts in degree at least six by lemma 3.4. They cannot change the quartic coefficient.

For $k = 3$, the leading complementary gradient is the degree-two vector

$$g = 3(a^2, b^2, a^2, b^2).$$

It lies in the period-two complement of the kernel. On a complementary vector (c, d, c, d) , the Hessian acts as $(2d, 2c, 2d, 2c)$. The leading elimination correction is $-H_4^{-1}g$, and its quartic change of the potential is obtained by substituting into the quadratic and linear terms:

$$-\frac{1}{2}g^T H_4^{-1}g = -9a^2 b^2. \quad (5.6)$$

Indeed $H_4^{-1}g = \frac{3}{2}(b^2, a^2, b^2, a^2)$. The restricted cubic unary term is zero by the opposite kernel pairs, and all other terms of this expansion have degree above four. Adding the pair coefficient yields $\alpha = 1 - 9 = -8$.

5.3 The pure-axis coefficient

On the axis $b = 0$, the first complementary correction is

$$u_0 = (0, c, 0, c), \quad c = -\frac{k}{2}a^{k-1}. \quad (5.7)$$

The leading unary gradient is ka^{k-1} at the two a -sites, and the Hessian of u_0 contributes $2c$ there, so these terms cancel. This vector is orthogonal to both kernel basis vectors. As in the path calculation, the higher pair gradients have orders at least $k + 1$ and $2k - 1$, while the corrected unary gradients have order $(k - 1)^2$. The inequalities (4.4) apply. Hence the complementary residual has order at least $k + 1$, and replacing u_0 by the exact solution changes the potential only in degree at least $2k + 2$.

At $(a, c, -a, c)$ the four pair terms sum to

$$2 \operatorname{Li}_2(ac) + 2 \operatorname{Li}_2(-ac) = \operatorname{Li}_2(a^2 c^2),$$

with degree- $2k$ coefficient $k^2/4$. The opposite a -sites contribute $-\frac{1}{2} \operatorname{Li}_2(a^{2k})$, with coefficient $-1/2$. The two c -sites have leading total unary term $2c^k$. This is of degree above $2k$ for $k \geq 5$, and equals $-(27/4)a^6$ for $k = 3$. Thus

$$\beta = \frac{k^2}{4} - \frac{1}{2} \quad (k \geq 5), \quad \beta = \frac{9}{4} - \frac{1}{2} - \frac{27}{4} = -5 \quad (k = 3).$$

Each paired expression has first degree $2k$, and the corrected unary degree is at least $2k$. Together with the error estimate and separate evenness, this proves (5.4) including its pure-axis order.

5.4 Four finite intersections

In terms of W in (5.3), the coefficients give

$$\begin{aligned} W(A, 0) &= \beta A^k + O(A^{k+1}), & W(0, B) &= \beta B^k + O(B^{k+1}), \\ W(A, B) &= \alpha AB + \text{terms of total degree at least three.} \end{aligned} \quad (5.8)$$

The absence of degree-two pure terms uses $k \geq 3$. The Jacobian of (W_A, W_B) at zero is $\begin{pmatrix} 0 & \alpha \\ \alpha & 0 \end{pmatrix}$, which is invertible. Formal inversion implies

$$(W_A, W_B) = (A, B) \quad \text{in } \mathbb{C}[[A, B]].$$

After substitution $A = a^2$, $B = b^2$, the ideal generated by these two derivatives is therefore (a^2, b^2) and has colength four.

The two effective gradient components are $2aW_A$ and $2bW_B$. For two series in $\mathbb{C}[[a, b]]$ with finite common colength, write $I(f, g) = \text{length } \mathbb{C}[[a, b]]/(f, g)$. The four choices of factors have finite intersections: the coordinate intersection has length one, the two axis intersections have length $2k - 2$ by (5.8), and the derivative intersection has length four as just shown. Applying lemma 3.3 gives

$$\begin{aligned} \mu(p_4) &= I(a, b) + I(a, W_B) + I(W_A, b) + I(W_A, W_B) \\ &= 1 + (2k - 2) + (2k - 2) + 4 = 4k + 1. \end{aligned} \quad (5.9)$$

Lemma 3.4 identifies this with $\mu(V_4)$, and (5.2) gives the same value for every $m = 4r$. This proves proposition 5.1 without a Newton nondegeneracy assumption.

Completion of the proof of theorem 2.2. Propositions 4.1 and 5.1 prove that every complete factor choice in (3.7) has finite colength and give its exact value. Lemma 3.3 therefore applies and gives (2.4). The common local presentation and the phase isomorphism are proposition 3.1. Finally multiply the common length by the support counts in lemma 2.1. $\square$

6 Rational multiplicity-generating functions

We now aggregate the local formula without introducing an orbit-counting interpretation. Let

$$P_m = \sum_{S \subseteq C_m} \prod_{R \in \text{Runs}(S)} w_{|R|}(k)$$

be its proper-subset part. Set $w_0 = 1$ for a block with no selected site. The periodic path weights give

$$R_k(t) := \sum_{L \geq 1} w_L(k) t^L = \frac{(k-1)t + t^2 + (2k-1)t^3 + t^4}{1 - t^4}. \quad (6.1)$$

A separator followed by a possibly empty selected run has series

$$B_k(t) = t(1 + R_k(t)) = \frac{t + (k-1)t^2 + t^3 + (2k-1)t^4}{1-t^4}. \quad (6.2)$$

For fixed m, q , let $P_{m,q}$ be the total weight of subsets with exactly $q \geq 1$ unselected sites. Marking one of these sites yields weighted count $qP_{m,q}$. Alternatively choose the mark's labeled position in m ways and read the cyclic word from that site. There are q ordered blocks, each a separator followed by a selected run. Their total weighted count is $m[t^m]B_k(t)^q$. Thus

$$qP_{m,q} = m[t^m]B_k(t)^q. \quad (6.3)$$

This equality holds also for rotationally symmetric words: sites are labeled, and no quotient by rotation has been taken. The empty selected subset corresponds to $q = m$ with all runs empty, so it is included.

Since B_k has zero constant term, the following sums are defined coefficientwise in $\mathbb{C}[[t]]$. Summing (6.3) gives

$$\sum_{m \geq 1} \frac{P_m}{m} t^m = \sum_{q \geq 1} \frac{B_k(t)^q}{q} = -\log(1 - B_k(t)). \quad (6.4)$$

By (2.5) and (6.2),

$$1 - B_k(t) = \frac{D_k(t)}{1-t^4}.$$

Define T_m exactly by the formal coefficients

$$\sum_{m \geq 1} \frac{T_m}{m} t^m = -\log D_k(t). \quad (6.5)$$

Equations (6.4) and (6.5) imply

$$P_m = T_m - 4\mathbf{1}_{4|m}.$$

Adding the full-cycle contribution from proposition 5.1 yields the useful alternative form

$$\ell_m(k) = T_m + 1 + 4(k-1)\mathbf{1}_{4|m}. \quad (6.6)$$

Exponentiating its formal sum divided by m proves (2.6). In particular (2.8) follows directly from the first four coefficients of $-\log D_k$; it is not an interpolation rule.

There is also a finite matrix expression for T_m , useful for exact algebraic checks. The companion matrix

$$M_k = \begin{pmatrix} 1 & k-1 & 1 & 2k \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad (6.7)$$

satisfies $\det(I - tM_k) = D_k(t)$, so the formal determinant identity gives $T_m = \text{tr}(M_k^m)$. For example, this identity follows by extending scalars to split the characteristic polynomial and expanding $-\log(1 - \lambda t)$ for each eigenvalue, counted with algebraic multiplicity; both sides are then identities in the original coefficient ring. This matrix represents the combinatorial coefficient formula. We do not assert that it realizes the local algebra as a finite dynamical system.

At the original clock n , lemma 2.1 gives

$$b_{2m} = [k+1 + (k-1)(-1)^m] \ell_m(k), \quad b_{2m-1} = 0. \quad (6.8)$$

Indeed the bracket is 2 for odd m and $2k$ for even m . Substitution into its formal exponential gives

$$\sum_{n \geq 1} \frac{b_n}{n} u^n = \frac{k+1}{2} \log Z_k(u^2) + \frac{k-1}{2} \log Z_k(-u^2),$$

which proves (2.7) and completes corollary 2.3. The exponents are integers because k is odd, so the native-clock expression is rational as well. Both series encode scheme lengths on the specified stratum, not ordinary periodic-point numbers.

7 Finite controls and limitations

7.1 What the exact checks verify

The proof above is uniform in k and in the number of sites. We also record bounded exact controls to make the delicate coefficients and the relation to the original equations independently testable. These computations are supplementary evidence; they are not used to extend a finite sample to the theorem’s quantified conclusion.

An exact symbolic calculation of the truncated potentials at the first complementary corrections gave table 1. For each tested odd k , the path leading degree and cycle-axis leading degree were $2k$, and every full-gradient residual had order at least $k+1$. The path columns show the coefficient of a^{2k} ; α is the mixed quartic coefficient and β the cycle-axis coefficient.

Table 1: Completed exact coefficient controls. These are checks of truncated formal substitutions, not original-equation local-length computations.

k	$L = 3$	$L = 7$	$L = 11$	Residual lower bound	α	β
3	−23	−46	−69	4	−8	−5
5	12	24	36	6	1	23/4
7	24	48	72	8	1	47/4

A separate symbolic enumeration of proper subsets of the labeled cyclic set for $1 \leq m \leq 8$, with k symbolic, checked (2.4) against the companion-trace expression (6.6). This is a finite Boolean-word enumeration, not a periodic-solution census. The all- m argument is the marked-separator identity (6.3).

An independent current-team check used the original polynomial equations (2.1), shifted to the point $(0, -1, 0, -1, \dots)$, with exact rational arithmetic and a local standard basis in Singular 4.2.1. It did not use the logarithmic presentation or the effective-potential coefficients. Its six completed local quotient lengths were those in table 2, all agreeing with (2.8).

Table 2: Completed independent local standard-basis controls from the original exchange equations.

k	$m = 1$	$m = 2$	$m = 3$
3	2	6	11
5	2	10	17

An initial direct attempt at $k = 3, m = 4$ was terminated after more than 77 seconds at approximately 0.9 GiB resident memory, without producing a local length. It is not a completed or passed check. The two-dimensional resonance in the theorem rests on the proof in section 5, not on a claimed standard-basis output for that case.

The accompanying research files preserve the exact coefficient/word script `check_effective_coefficients.py`, its bounded-check record `COEFFICIENT_CHECKS.md`, and the independently

authored `ROOT_CLUSTER_CHECK.sing` and `ROOT_CLUSTER_REVIEW.md` in the parent research directory. The scripts respectively reproduce the symbolic controls and the six completed local-equation cases. The latter excludes the terminated case. No claim of external or human peer review is attached to these controls.

7.2 The boundary that remains unclassified

The theorem concerns exactly the alternating-zero stratum. It neither exhausts nor declares absent the other possible zero supports in (2.1). One elementary relation indicates where a broader boundary analysis would require more information. Multiplying the relations at $i - 1$ and $i + 1$, and then using the relation at i , gives the identity in $\mathcal{A}_{k,n}$

$$x_i \left[-x_{i-2} - x_{i+2} + x_i x_{i-2} x_{i+2} - \frac{(1 + x_i^k)^k - 1}{x_i} \right] = 0. \quad (7.1)$$

The displayed quotient is a polynomial, since its numerator is divisible by x_i^k . At a point where $x_i = 0$ and $x_{i-2} + x_{i+2} \neq 0$, the bracket is a unit in the local algebra, so x_i vanishes in that local algebra, not just at the reduced point. When the neighboring sum is zero, this unit argument fails. Longer nonzero runs can therefore present additional local return problems not resolved by the path decomposition at an alternating point.

No total torus-localization length is obtained by subtracting only (6.8) from k^n . Nor has a comparison been constructed between the unsaturated local algebra and a saturated fixed scheme on a cluster surface. For $m \geq 3$ the embedding-dimension argument in remark 3.2 explicitly precludes the smooth-surface fixed-scheme identification. Even for the smaller values of m , where that particular obstruction does not apply, no such identification is asserted.

The proof uses characteristic zero in its formal logarithms, fractional powers, nonzero leading coefficients, and elimination steps. Oddness of k is used in both the support restriction and the cancellations of the effective potentials. Results for even exponents or positive characteristic require separate arguments. Finally, the series (2.6) and (2.7) have been defined using these local scheme multiplicities; they are not claimed to be Artin–Mazur zeta functions of F_k , of a smooth cluster-surface automorphism, or of another constructed dynamical system.

Within these boundaries the conclusion is complete: every alternating-zero local algebra, at every admissible clock and every odd exponent at least three, has the explicit length (2.4). Its two degenerate Hessian types account for the nontrivial weights, and their exact cyclic aggregation gives the rational local-multiplicity series.

References

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